GRAINGUARD · RISK REGISTER

GrainGuard risk register, hazard analysis and design FMEA

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First issue (kit 1.8 upgrade, stage P3): project risks, preliminary hazard analysis and a first-pass design FMEA

GrainGuard risk register, hazard analysis and design FMEA

The top project risk is that interstitial RH does not predict moisture well enough for the decision rule (PR-002), which a trial against oven samples settles at TRL 4. The top hazards are grain engulfment and falls while the cable is fitted, an automatic fan start during service, and the roof hanger overloaded when the bin is unloaded; the design controls each by procedure and by keeping GrainGuard's own parts at 12 V, but none can be designed out of a grain bin. The highest design FMEA numbers are the pod's moisture reading (RPN 180) and the battery in deep cold (RPN 160), the latter waiting on Amish's decision (GGD-DEC-001 open item 11). Method: STANDARDS section 23. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high); detection 1 (certain to be found) to 5 (unlikely to be found), and the FMEA uses a 1 to 10 scale for S, O and D.

> **Safety:** Grain bins kill people. This document lists hazards so they can be controlled; it does not make any task safe. The safety notes in GGD-PRC-001 and the safety stops in GGD-BLD-001 govern every task.

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	The moisture estimate misses ±0.8 points	Equation fit error, hysteresis, drift and membrane lag not quantified (STEM check S8)	Fan decisions off by about a point; PR-002 not met	3	4	Design lead	Compare pod readings with oven samples on the host farm at TRL 4; fit crop constants per lot if needed
P2	No host farm and electrician found	The route through an extension specialist (decided 2026-10-02) not yet started	No co-design visit; targets stay unvalidated	3	3	Project lead	Contact a land-grant grain-storage specialist when the cap lifts
P3	Bin makers will not rate their hangers for a cable	Liability; no data for a thin cable	The kit cannot be hung in many bins (PR-010)	3	4	Project lead	Ask the host bin's maker first; option B hangs almost no load (GGD-OPT-001)
P4	Cost stays well over the value-engineering target	Mast, pods and relay kit; prototype quantities	Weak case against commercial systems at the low end	4	2	Design lead	Savings in GGD-DEC-001 (Value engineering); the 24 V graft from option C
P5	The concept choice changes the design	Options within 10 % (GGD-DDR-004)	Rework of PRS, BOM, plan and drawings if B or C is picked	2	3	Project lead	Amish decides open item 10 before more work on option A
P6	Hazardous-location rules apply inside the bin	Grain dust is a combustible dust (GGD-REG-001 A2)	Pods and cable need an assessed design	2	4	Electronics lead	Assess with a reviewer before any product claim
P7	Battery freezes in a northern winter	Disconnect at the 50 % figure (S17)	Cracked battery, loss of control in the coldest weeks	3	3	Design lead	Amish decides open item 11 (recommended: the 65 % figure)

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Engulfment and entrapment in grain	Entering a bin holding grain to fit or service the cable	Death by suffocation	5	2	The cable is fitted only in an empty bin, with unloading locked out, an entry permit and a trained person outside (BLD S2); no task needs entry into a bin holding grain (PR-011)	5 x 1: procedure only; the rule must be kept
H2	Fall from the bin roof	Feeding the cable and fitting the peak-cap gland	Serious injury or death	5	2	Fall protection, ladder cage and roof rail, dry calm day (BLD S3)	5 x 1
H3	Automatic fan start	The controller starts the fan while someone works at the fan or transition	Injury from the impeller or plenum	4	2	Lock out at the disconnect; "Starts automatically" labels; the hand-off-auto switch is never a safety isolator (PRC, Safety)	4 x 1
H4	Electric shock at the starter	Fitting the relay kit in a mains starter	Death or burns	5	2	Licensed electrician, starter isolated and locked out (BLD S6); 2.5 kV isolation; GrainGuard's own parts at 12 V (PR-012)	5 x 1
H5	Roof overload	Grain dragging the cable during unloading	Roof buckling, falling parts	4	2	Hang only from a hanger the bin maker rates for cable loads; unload from the center (PR-010)	4 x 2 until the rating is known
H6	Battery hydrogen ignites	Gas collecting in the closed box during cold charging	Box burst, burns	3	2	Two labyrinth vents low and high (D10, 1.7 times the IEC 62485-2 opening); charge clamped at 15.0 V; fuse at the terminal	3 x 1
H7	Battery short circuit	Wiring error or damaged lead	Burns, fire	3	2	5 A fuse at the battery positive; fuse out until the checks (BLD)	3 x 1
H8	Mast falls in a storm	Stay left off; anchors in weak concrete	Injury, damage	4	2	Stay off only in gusts under 30 m/s and refitted before leaving (D12); anchors to the maker's data (S26)	4 x 1
H9	Phosphine exposure or false reassurance	Someone reads GrainGuard as a sign that a fumigated bin is safe	Poisoning	5	1	GrainGuard does not measure phosphine and says so (PRC, Safety; REG R7); only certified applicators fumigate	5 x 1

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H10	Grain dust ignition	A fault in the pods or cable in a dusty headspace	Dust fire or explosion	5	1	12 V and a few milliamps; sealed glands; hazardous-location assessment to come (REG A2)	5 x 1, unassessed
H11	Zinc fumes when welding the mast	Welding galvanized pipe	Metal fume fever	2	3	Grind the zinc off first; weld in open air (BLD 3.4)	2 x 1
H12	Lightning surge through the cable	Tall steel bin; cable from the roof	Damage to the controller; fire risk	3	2	Bond rope and shield to the bin; surge protector on the bus (BOM line 16)	3 x 1

Design FMEA (first pass, P3)

Table 3. Failure modes. RPN = severity x occurrence x detection, each scored 1 to 10.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Sensor pod (BOM 2)	Read grain T and interstitial RH	RH reading biased	Wrong moisture estimate; fan runs or stays off wrongly	Membrane fouled with fines; sensor drift; condensation in the pod	6	5	6	180	Compare with oven samples at TRL 4; drift check against the ambient sensor with the fan running; replaceable membrane
F2	Battery (BOM 6) and disconnect (BOM 7)	Store energy through winter	Electrolyte freezes and the case cracks	Station dead in the coldest weeks; acid inside the box	Disconnect at the 50 % figure, -30 °C night (S17)	8	5	4	160	Open item 11: the 65 % figure; confirm with the AGM maker's data at TRL 4
F3	Bus cable and glands (BOM 3)	Carry 12 V and RS-485	Cable cut or pulled out of a pod gland	Pods below the break go silent	Grain drag during unloading; rodents	6	4	3	72	Jumpers tied to the rope every 300 mm; stale-data alarm; strain relief in the glands
F4	Plenum probe (BOM 14)	Measure the fan heat	Probe reads high or low	Plenum EMC off by about one point per degree	Probe out of the air stream; sheath fouled; below -10 °C its accuracy falls	6	3	5	90	Check the probe against the ambient sensor with the fan off (TRL 4); insertion depth set in the plan (48 mm)
F5	Relay kit (BOM 11)	Switch the fan	Relay welded shut	Fan runs when it should not	Contact wear; inrush	6	2	3	36	Current transformer confirms fan state; mismatch alarm (PR-005)
F6	Relay kit (BOM 11)	Fail off	Fan keeps running after controller power loss	Rewetting or rewarming the grain	Wrong relay type (normally closed)	7	2	2	28	Normally open relay specified; first check in the plan (PR-005)
F7	Solar panel (BOM 8)	Recharge the battery	Too little winter energy	Battery runs down over weeks	Mast on the north side, shaded by the bin; snow on the panel	7	3	4	84	Mast south of the bin (D13); 45 degree tilt sheds snow; low-battery alarm
F8		Keep electronics	Water enters		Driven rain into the labyrinth	6	3	5	90	Labyrinth vents with their

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
	Enclosure and vents (BOM 4)	dry and vent hydrogen	through a vent	Corroded board; short circuit	vents (IP55 class)					openings facing down; confirm their rating from the maker; inspect at the first service
F9	Mast and stay (BOM 10)	Hold panel, box, shield and antenna	Mast yields in a gust	Panel and box fall	Stay left off in a storm; S195T tube	8	2	3	48	Service rule (D12); check the stay is fitted at every visit
F10	Rope and top fittings (BOM 1)	Carry the pods	Rope or hanger fails	Cable and pods fall into the grain; roof damage	Unrated hanger; corroded clips	8	2	6	96	Hang only from a rated hanger; inspect the clips at each empty-bin visit
F11	Radio (BOM 5, 15)	Report to the farmhouse	Packets lost	No alarm at the house	Buildings and the bin in the path	5	4	3	60	Site survey at TRL 4; retries; stale-data alarm at the receiver
F12	Pod boards (BOM 2)	Survive fumigation	Copper tracks corrode	Pods fail after a fumigation	Phosphine (S28)	6	4	5	120	Coated test board through one fumigation at TRL 4; sealed connectors if it corrodes